

1. Questions

Vishnu has Rs x . He invested 30% of Rs x in scheme A which offers $r\%$ annual simple interest, and the remaining amount in scheme B which offers $(r+3)\%$ compound interest which is compounded annually. At the end of two years, the total interest received by Vishnu from both schemes is Rs 1512 and the difference between interest received from both schemes is Rs 840. What will be the amount when Rs x is invested in scheme B for three years? (value of r is a single digit number)

- a. Rs. 11458
- b. Rs. 10648
- c. Rs. 9576
- d. Rs. 10289
- e. None of these

2. Questions

A, B and C together can complete a work in y days. The efficiency of B is 60% of the efficiency of C and 50% of the efficiency of A. A alone can complete 40% of the total work in $(y-1)$ days. A total wages of Rs 1680 is paid to complete of the work. If B and C (in the same order) started working alternatively, then find the amount earned by B out of the total wages to complete the work.

- a. Rs. 640
- b. Rs. 650
- c. Rs. 720
- d. Rs. 560
- e. None of these

3. Questions

P, Q and R jointly started a business by investing Rs $1.5x$, Rs $(x+2000)$ and Rs $1.7x$ respectively. At the end of one year, both P and Q withdrew Rs 500 and Rs 2000 respectively. At the end of three years, out of the total profit of Rs 20000, 32% profit is stolen by someone. The amount of profit received by Q is Rs 5000. What is the average of the initial investment of Q and R?

- a. Rs. 2500
- b. Rs. 2750
- c. Rs. 3175
- d. Rs. 2350
- e. None of these

4. Questions

Train M can cross x m of platform in 40 seconds. The length of train L is x m less than that of train

M. The ratio of the speeds of trains L and M is 5:4. Train L can overtake train M in 200 seconds. The length of train M is $(4x - 200)$ m. Train L can cross a bridge (which is twice the length of train L) in t seconds. What is the value of $(x+t)$?

- a. 212
- b. 340
- c. 248
- d. 168
- e. None of these

5. Questions

A shopkeeper sold two articles P and Q at the same price. The profit percent on P and Q is 12% and 25% respectively. The profit percent of article P is calculated on cost price and the profit percent of article Q is calculated on cost price. Total cost price of both articles is Rs 4740. The selling price of article Q is how much more than the cost price of article P?

- a. Rs. 250
- b. Rs. 300
- c. Rs. 290
- d. Rs. 320
- e. None of these

6. Questions

In each question a number series is given. Read the given information carefully and answer the questions based on it after solving.

The given number series contains a wrong term.

Series: 2330, P, 2240, Q, 2087, 1985

$$P - Q = 17 \times 17 - 11 \times 11 - 49$$

$$Q = 13^3 - 5^2$$

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A new series of numbers begins with the fourth term of the above series and follows the same pattern as the given series. What will be the average of the third terms of both series?

- a. 2163.5
- b. 2250
- c. 3461
- d. 1836
- e. None of these

7. Questions

Each series contains a missing term.

Series 1: 81, 649, 2597, 5195, ____, 2599

Series 2: 251, ____, 216, 190, 157, 116

A and B are the missing terms of series 1 and series 2 respectively. What will be the remainder when A is divided by B?

- a. 3
- b. 11
- c. 4
- d. 7
- e. None of these

8. Questions

Each of the given series contains a wrong term.

Series 1: 50, 90, 164, 272, 422, 620

Series 2: 845, 1070, 814, 1103, 779, 1150

Which of the given statements is or are true?

- I). Second last term of the series 2 is wrong.
 - II). The average of correct term of both series that will replace the wrong term is 616.
 - III). Difference between the wrong terms of both series is 1050.
- a. Only I
 - b. Only III
 - c. Both I and II
 - d. All three
 - e. None of these

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9. Questions

Series 1 contains a wrong term.

Series 1: 5760, 960, 192, 48, 19, 8

Series 2: 662, $(10A + 18)$, 146, 60, B

Last term of series 2 is a prime number between wrong term and correct term of series 1.

What is the quotient after dividing the first term in the series 1 by A?

- a. 175
- b. 126
- c. 154
- d. 192
- e. None of these

10. Questions

Each of the given series contains a wrong term.

Series 1: 960, 964, 995, 1007, 1132, 1168

Series 2: 30, 15, 15, 22.5, 52.5, 112.5

The first term of series 3 is equal to the correct term that will replace the wrong term of series 2 and series 3 follows the same pattern as series 1. What is the average of 5th and 6th terms of series 3?

- a. 235
- b. 515
- c. 397.5
- d. 234.25
- e. None of these

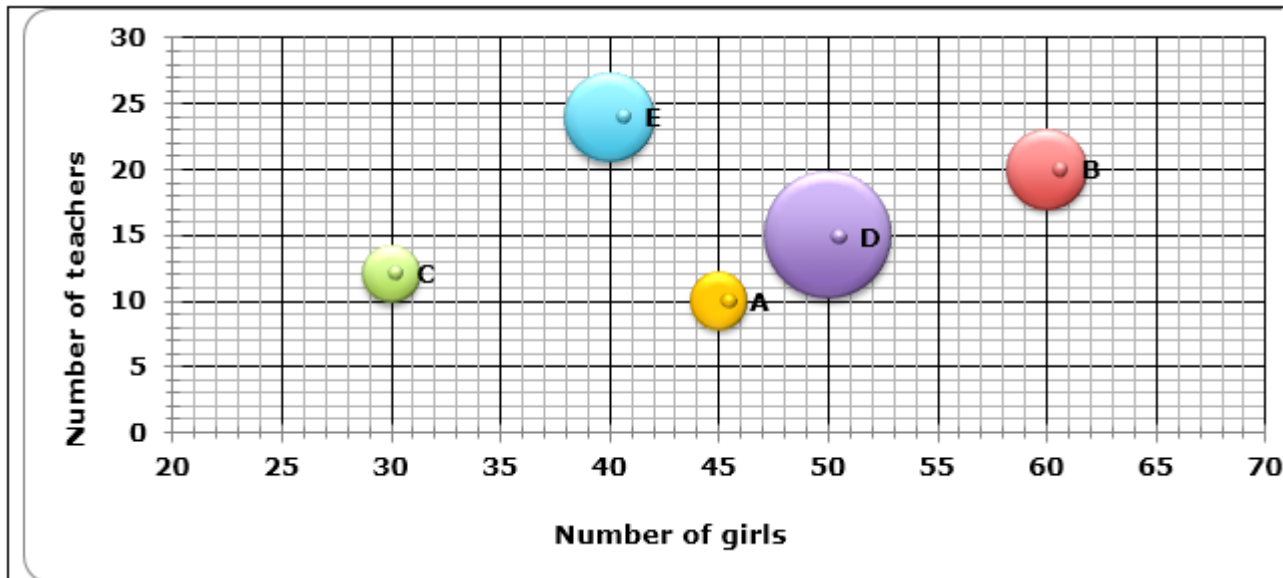
11. Questions

Read the data carefully and answer the given questions based on it.

The given bubble chart represents the number of students and teachers participating in an International conference from five different schools of India.

x-axis shows the number of girls and y-axis shows the number of teachers of five different schools.

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Notes:

- i). The ratio of the number of students in schools A, B, and C is 21:30:16, and the total number of boys in these three schools is 200.
- ii). The number of boys in School D is 3 less than twice the number of teachers in School E.
- iii). The number of boys in School E is y less than the total number of girls in Schools A and B combined, and the number of girls in School E is 28.57% of the total number of students in School E.

If a group of 5 students is to be made and one teacher is to lead each group, then find how many more teachers would be needed for all the students of school C?

- a. $2y$
- b. y
- c. $(y-1)$
- d. $(y+3)$
- e. None of these

12. Questions

Establish the relation between given quantities:

QI: The difference between the sum of the number of teachers and students of school B and that of school D.

QII: The difference between the number of boys from all five schools together and the number of girls from all five schools together.

- a. $QI > QII$
- b. $QI < QII$
- c. $QI \geq QII$
- d. $QI \leq QII$

e. $QI = QII$

13. Questions

Which of the following statements is true?

I: In School B, boys make up 60% of the total number of students.

II: The average number of teachers in the five schools is 20.

III: The ratio of the number of students in schools C and E is 4:7.

- a. Only I
- b. Only II
- c. All three
- d. Both I and III
- e. None of these

14. Questions

There are a total of 30y students and teachers from School X attending the conference. The number of teachers in school X is the average of the number of teachers in schools B and C combined. The number of students in school X is what percentage more or less than the number of girls in schools B and E combined?

- a. $7y\%$
- b. $(6y+4)\%$
- c. $(y+20)\%$
- d. $(7y-4)\%$
- e. None of these

15. Questions

In school D, the number of girls is 'a' more than the number of boys. In school C, the number of boys is 'b'% of the total number of students. What is the value of $(2b-a)$?

- a. 100
- b. 75
- c. 110
- d. 150
- e. None of these

16. Questions

Read the given information carefully and find the answer based on it.

I: $3x^2 + (P+10)x + 63 = 0$

II: $y^2 - Qy = (P+4)$

Q=P/10. The sum of the roots of equation I is -10. Find the relationship between the roots of the two equations:

- a. $x > y$
- b. $x < y$
- c. $x \geq y$
- d. $x \leq y$
- e. $x = y$ or relation cannot be established

17. Questions

Read the given information carefully and find the answer based on it.

I: $x^2 - Mx - 3MN = 0$

II: $7y^2 + (M-1)y - 3M = 0$

The product of the roots of equation I is -180. One of the roots of equation II is 1. Find the sum of the smaller roots of both equations?

- a. $-84/11$
- b. -11
- c. $-93/7$
- d. $-122/7$
- e. None of these

18. Questions

Read the given information carefully and find the answer based on it.

A and B are two natural numbers. The LCM and HCF of A and B are 38 and 19, respectively. ($A > B$)

I: $x^2 - Ax + 105 = 0$

II: $y^2 - 9y + (B-1) = 0$

Establish the relationship between the roots of the two equations:

- a. $x > y$
- b. $x < y$
- c. $x \geq y$

- d. $x \leq y$
- e. $x = y$ or relation cannot be established

19. Questions

Read the given information carefully and find the answer based on it.

I: $x^2 - (K-1)x + (5K+2) = 0$

II: $y^2 - 25y + D^2 = 0$

One of the roots of equation I is the square of the smallest odd prime number. $D = (K-11)$. What is the sum of the positive roots of both equations?

- a. 54
- b. 47
- c. 64
- d. 76
- e. None of these

20. Questions

Read the given information carefully and find the answer based on it.

Column A	Column B
I: $5x^3 - 24x^2 - 77x = 0$	p: Both the roots are negative.
II: $7y^2 + 61y + 54 = 0$	q: One of its roots is prime number.
III: $2z^2 - 9z - 45 = 0$	r: Both the roots are positive values.

Choose the correct answer from the given options:

- a. I-p, II-q, III-r
- b. I-q, II-p, III-r
- c. I-q, II-p
- d. II-p, III-p
- e. None of these

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Explanations:

1. Questions

Answer: B

Total amount of Vishnu = Rs x = Rs 100a

SI = r%

CI = (r+3)%

He invested 30% of Rs x in scheme A which offers r% annual simple interest and remaining amount in scheme B which offers (r+3)% compound interest which is compounded annually.

Total interest received by Vishnu from both schemes after two years = Rs 1512

$$30a * 2 * r\% + 70a * [(100 + r + 3)/100]^2 - 70a = 1512$$

$$(3a * r/5) + 70a * [(103^2 + r^2 + 206r - 100^2)/100^2] = 1512$$

$$(3a * r/5) + 70a * [(609 + r^2 + 206r)/100^2] = 1512 \dots \dots \dots (1)$$

Difference between interest received from both schemes = Rs 840

$$70a * [(609 + r^2 + 206r)/100^2] - (3a * r/5) = 840 \dots \dots \dots (2)$$

$$(1) - (2)$$

$$2 * (3a * r/5) = (1512 - 840)$$

$$6a * r/5 = 672$$

$$a * r = 560 \rightarrow (3)$$

$$(1) + (2)$$

$$2 * 70a * [(609 + r^2 + 206r)/100^2] = 1512 + 840$$

$$a * (609 + r^2 + 206r) = 168000$$

$$609a + ar^2 = 206ar = 168000$$

$$\text{Put } (ar = 560)$$

$$609a + 560r = 168000 - 115360$$

$$609 * (560/r) + 560r = 52640$$

$$(609/r) + r = 94$$

$$609 + r^2 = 94r$$

$$r^2 - 94r + 609 = 0$$

$$r^2 - 87r - 7r + 609 = 0$$

$$(r - 87) (r - 7) = 0$$

Roots of this equation are 87 and 7

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$$r = 7$$

$$a = 80$$

Total amount = Rs 8000

Compound interest = 10%

Required answer = $8000 \times 133.1\% = 10648$ Rs

Hence the answer is option B.

2. Questions

Answer: A

Time taken by A, B and C together to complete a work = y days

Time taken by A to complete the work = $5(y-1)/2$ days

Ratio of the efficiency of A, B and C = 6:3:5

$$y \times 14 = 6 \times 5(y-1)/2$$

$$y = 15$$

Time taken by A to complete the work = 35 days

Let the total work = 210 unit

Total wages of = Rs 1680

B and C (in the same order) alternatively completed the work

Amount of work completed by B = $26 \times 3 + 2 = 80$ unit

Required answer = $1680 \times 80 / 210 =$ Rs 640

Hence the answer is option A.

3. Questions

Answer: D

Ratio of the profit share of P, Q and R

$$= [1.5x \times 1 + (1.5x - 500) \times 2] : [(x + 2000) \times 1 + x \times 2] : 1.7x \times 3$$

$$= (4.5x - 1000) : (3x + 2000) : 5.1x$$

Total profit = Rs 20000

32% profit is stolen by someone.

Profit amount received by Q = Rs 5000

$$20000 \times 68\% \times (3x + 2000) / (12.6x + 1000) = 5000$$

$$315x + 25000 = 204x = 136000$$

$$111x = 111000$$

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$$x = 1000$$

$$\text{Required answer} = (3000 + 1700)/2 = \text{Rs } 2350$$

Hence the answer is option D.

4. Questions

Answer: C

$$\text{Length of train M} = (4x - 200) \text{ m}$$

$$\text{Length of train L} = (3x - 200) \text{ m}$$

$$\text{Let the speed of train L} = 5a \text{ mps}$$

$$\text{Speed of train M} = 4a \text{ mps}$$

Train M can cross x m platform in 40 sec.

$$(4x - 200) + x = 4a * 40$$

$$5x - 160a = 200$$

$$x - 32a = 40 \dots\dots\dots(1)$$

Train L can overtake train M in 200 sec.

$$(4x - 200) + (3x - 200) = 200 * a$$

$$7x - 200a = 400 \dots\dots\dots(2)$$

Solving both equations

We get

$$x = 200$$

$$a = 5$$

$$\text{Length of train L} = 400 \text{ m}$$

$$\text{Speed of train L} = 25 \text{ mps}$$

Train L can cross A bridge (having double length of the train L) in t sec.

$$t = (400 + 800)/25 = 48 \text{ sec}$$

$$\text{Value of } (x+t) = 200 + 48 = 248$$

Hence the answer is option C.

5. Questions

Answer: B

$$\text{Let the selling price of article P and Q} = \text{Rs } 140a$$

$$\text{Cost price of article P} = \text{Rs } 125a$$

$$\text{Cost price of article Q} = \text{Rs } 112a$$

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Total cost price of both articles = Rs 4740

$$125a + 112a = 4740$$

$$a = 20$$

$$\text{Required answer} = (140 - 125) \times 20 = \text{Rs } 300$$

Hence the answer is option B.

6. Questions

Answer: A

$$Q = 13^3 - 5^2$$

$$Q = 2197 - 25$$

$$Q = 2172$$

$$P - Q = 17 \times 17 - 11 \times 11 - 49$$

$$P - 2172 = 289 - 121 - 49$$

$$P = 2291$$

2330	2291	2240	2172	2087	1985
	-39	-51	-68	-85	-102

Pattern: -34, -51, -68, -85, -102

Wrong term = 2330

$$\text{Correct term} = 2291 + 34 = 2325$$

New series: 2172, 2138, 2087, 2019, 1934, 1832

$$\text{Required answer} = (2240 + 2087) / 2 = 2163.5$$

Hence the answer is option A.

7. Questions

Answer: C

Series 1: 81, 649, 2597, 5195, ____, 2599

81	649	2597	5195	5196	2599
	*8+1	*4+1	*2+1	*1+1	*0.5+1

Missing term = 5196

Series 2:

251	236	216	190	157	116
	15	20	26	33	41
	5	6	7	8	

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Missing term = 236

A = 5196

B = 236

Required answer = 4

Hence the answer is option C.

8. Questions

Answer: E

Series 1:

50	92	164	272	422	620
	+42	+72	+108	+150	+198
		+30	+36	+42	+48

Wrong term = 90

Correct term = 92

Series 2:

845	1070	814	1103	779	1150
	+225	-256	+289	-324	+371
	15^2	16^2	17^2	18^2	19^2

Wrong term = 1150

Correct term = 1140

I). Second last term of the series 2 is wrong. (false)

II). Average of correct term of both series that will replace the wrong term = $(1140 + 92)/2 = 616$ (true)

III). Difference between the wrong terms of both series = $1150 - 90 = 1060$ (false)

Hence the answer is option E.

9. Questions

Answer: D

Series 1:

5760	960	192	48	16	8
	÷6	÷5	÷4	÷3	÷2

Wrong term = 19

Correct term = 16

Last term of series 2 is a prime number between wrong term and correct term of series 1.

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B = 17

Series 2:

662	318	146	60	17
	344	172	86	43
	÷2	÷2	÷2	

$$318 = 10A + 18$$

$$10A = 300$$

$$A = 30$$

Required answer = $5760/30 = 192$

Hence the answer is option D.

10. Questions

Answer: A

Series 1:

960	964	991	1007	1132	1168
	+2 ²	+3 ³	+4 ²	+5 ³	+6 ²

Wrong term = 995

Correct term = 991

Series 2:

30	15	15	22.5	45	112.5
	*0.5	*1	*1.5	*2	*2.5

Wrong term = 52.5

Correct term = 45

Series 3:

45	49	76	92	217	253
	+2 ²	+3 ³	+4 ²	+5 ³	+6 ²

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Required answer = $(253 + 217)/2 = 235$

Hence the answer is option A.

11. Questions

School	Number of teachers	Number of girls
A	10	45
B	20	60
C	12	30
D	15	50
E	24	40

The ratio of the number of students in school A, B and C together= 21a:30a:16a

Total number of boys in schools A, B and C together = 200

Total number of students in A, B and C together= 200+ 135 = 335

$$21a+30a+16a = 335$$

$$67a = 335$$

$$a = 5$$

Number of boys in school D = $2 \times 24 - 3 = 45$

Number of girls is 28.57% of total number of students in school E.

Ratio of the number of girls and boys in school E = 2:5

Number of boys in school E = 100

Number of boys of school E is y less than the total number of girls of school A and B together

$$y = 105 - 100$$

$$y = 5$$

School	Number of teachers	Number of girls	Number of boys	Total number of students
A	10	45	60	105
B	20	60	90	150
C	12	30	50	80
D	15	50	45	95
E	24	40	100	140

Answer: C

Number of groups of 5 students from school C = $80/5 = 16$

Number of teachers required = $16 - 12 = 4$

Hence the answer is option C.

12. Questions

School	Number of teachers	Number of girls
A	10	45
B	20	60
C	12	30
D	15	50
E	24	40

The ratio of the number of students in school A, B and C together= 21a:30a:16a

Total number of boys in schools A, B and C together = 200

Total number of students in A, B and C together= 200+ 135 = 335

$$21a+30a+16a = 335$$

$$67a = 335$$

$$a = 5$$

Number of boys in school D = $2 \times 24 - 3 = 45$

Number of girls is 28.57% of total number of students in school E.

Ratio of the number of girls and boys in school E = 2:5

Number of boys in school E = 100

Number of boys of school E is y less than the total number of girls of school A and B together

$$y = 105 - 100$$

$$y = 5$$

Answer: B

QI:

The difference between the sum of the number of teachers and students of school B and that of school D

$$= (150 + 20) - (15 + 95)$$

$$= (170 - 110) = 60$$

QII:

The difference between the number of boys from all five schools together and the number of girls from all five schools together

$$= (60 + 90 + 50 + 45 + 100) - (45 + 60 + 30 + 50 + 40)$$

$$= 345 - 225$$

$$= 120$$

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QI < QII

Hence the answer is option B.

13. Questions

School	Number of teachers	Number of girls
A	10	45
B	20	60
C	12	30
D	15	50
E	24	40

The ratio of the number of students in school A, B and C together= 21a:30a:16a

Total number of boys in schools A, B and C together = 200

Total number of students in A, B and C together= 200+ 135 = 335

$$21a+30a+16a = 335$$

$$67a = 335$$

$$a = 5$$

Number of boys in school D = $2 \times 24 - 3 = 45$

Number of girls is 28.57% of total number of students in school E.

Ratio of the number of girls and boys in school E = 2:5

Number of boys in school E = 100

Number of boys of school E is y less than the total number of girls of school A and B together

$$y = 105 - 100$$

$$y = 5$$

Answer: D

I: In School B, boys make up 60% of the total number of students.

$$\text{Required answer} = (90/150) \times 100 = 60\% \text{ (true)}$$

II: The average number of teachers in the five schools= $(10 + 20 + 12 + 15 + 24)/5 = 16.2$ (false)

III: The ratio of the number of students in schools C and E= $80:140 = 4:7$ (true)

Hence the answer is option D.

14. Questions

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School	Number of teachers	Number of girls
A	10	45
B	20	60
C	12	30
D	15	50
E	24	40

The ratio of the number of students in school A, B and C together = 21a:30a:16a

Total number of boys in schools A, B and C together = 200

Total number of students in A, B and C together = 200 + 135 = 335

$$21a + 30a + 16a = 335$$

$$67a = 335$$

$$a = 5$$

Number of boys in school D = $2 \times 24 - 3 = 45$

Number of girls is 28.57% of total number of students in school E.

Ratio of the number of girls and boys in school E = 2:5

Number of boys in school E = 100

Number of boys of school E is y less than the total number of girls of school A and B together

$$y = 105 - 100$$

$$y = 5$$

Answer: B

Total number of students and teachers from school X = 150

Number of teachers in school X = $(20 + 12)/2 = 16$

Number of students from school X = $150 - 16 = 134$

Total number of girls from schools B and E together = 100

Required answer = $[(134 - 100)/100] \times 100 = 34\%$

Hence the answer is option B.

15. Questions

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School	Number of teachers	Number of girls
A	10	45
B	20	60
C	12	30
D	15	50
E	24	40

The ratio of the number of students in school A, B and C together = $21a:30a:16a$

Total number of boys in schools A, B and C together = 200

Total number of students in A, B and C together = $200 + 135 = 335$

$$21a + 30a + 16a = 335$$

$$67a = 335$$

$$a = 5$$

Number of boys in school D = $2 \times 24 - 3 = 45$

Number of girls is 28.57% of total number of students in school E.

Ratio of the number of girls and boys in school E = 2:5

Number of boys in school E = 100

Number of boys of school E is y less than the total number of girls of school A and B together

$$y = 105 - 100$$

$$y = 5$$

Answer: E

In school D number of girls is 'a' more than the number of boys.

$$a = 50 - 45$$

$$a = 5$$

In school C number of boys is 'b'% of the total number of students.

$$b\% = (50/80) \times 100$$

$$b\% = 62.5\%$$

$$b = 62.5$$

$$\text{Value of } (2b - a) = (2 \times 62.5 - 5) = 120$$

Hence the answer is option E.

16. Questions

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Answer: E

I: $3x^2 + (P+10)x + 63 = 0$

Sum of roots of equations I = -10

$$[-(p + 10)/3] = -10$$

$$p + 10 = 30$$

$$P = 20$$

$$3x^2 + 30x + 63 = 0$$

$$x^2 + 10x + 21 = 0$$

$$x^2 + 7x + 3x + 21 = 0$$

$$(x + 7)(x + 3) = 0$$

Roots of equations are -7 and -3.

$$Q = 20/10 = 2$$

II: $y^2 - Qy = (P+4)$

$$y^2 - 2y - 24 = 0$$

$$y^2 - 6y + 4y - 24 = 0$$

$$(y - 6)(y + 4) = 0$$

Roots of equations are +6 and -4.

The value of 'x'	Relation	The value of 'y'
-7	<	+6
-7	<	-4
-3	<	+6
-3	>	-4

Relation cannot be established between the roots.

Hence the answer is option E.

17. Questions

Answer: C

II: $7y^2 + (M-1)y - 3M = 0$

One of the roots of equation II = 1

$$7 + (M-1) - 3M = 0$$

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$$M = 3$$

$$7y^2 + 2y - 9 = 0$$

$$7y^2 - 7y + 9y - 9 = 0$$

$$(7y + 9)(y - 1)$$

Roots of this equation are $-9/7$ and 1 .

$$\text{I: } x^2 - Mx - 3MN = 0$$

Product of roots of equation I = -180

$$3MN = 180$$

$$N = 20$$

$$x^2 - 3x - 180 = 0$$

$$x^2 - 15x + 12x - 180 = 0$$

$$(x - 15)(x + 12) = 0$$

Roots of this equation are $+15$ and -12

$$\text{Required answer} = -9/7 - 12 = -93/7$$

Hence the answer is option C.

18. Questions

Answer: E

LCM of A and B = 38

HCF of A and B = 19

$$A > B$$

$$A = 38$$

$$B = 19$$

$$\text{I: } x^2 - Ax + 105 = 0$$

$$x^2 - 38x + 105 = 0$$

$$x^2 - 35x - 3x + 105 = 0$$

$$(x - 35)(x - 3) = 0$$

Roots of equation are 35 and 3

$$\text{II: } y^2 - 9y + (B-1) = 0$$

$$y^2 - 9y + 18 = 0$$

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$$y^2 - 6y - 3y + 18 = 0$$

$$(y - 6)(y - 3) = 0$$

Roots of this equation are 6 and 3

The value of 'x'	Relation	The value of 'y'
+35	>	+6
+35	>	+3
+3	<	+6
+3	=	+3

Hence the answer is option E.

19. Questions

Answer: B

$$\text{I: } x^2 - (K-1)x + (5K+2) = 0$$

One of the roots of equation I = 9

$$81 - 9(K-1) + (5K+2) = 0$$

$$K = 23$$

$$x^2 - 22x + 117 = 0$$

$$x^2 - 13x - 9x + 117 = 0$$

$$(x - 13)(x - 9) = 0$$

Roots of this equation are 13 and 9

$$D = (K-11)$$

$$D = 12$$

$$\text{II: } y^2 - 25y + D^2 = 0$$

$$y^2 - 25y + 144 = 0$$

$$y^2 - 16y - 9y + 144 = 0$$

$$(y - 16)(y - 9) = 0$$

Roots of this equation are 16 and 9

$$\text{Required answer} = 13 + 9 + 16 + 9 = 47$$

Hence the answer is option B.

20. Questions

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Answer: C

I: $5x^3 - 24x^2 - 77x = 0$

$$x(5x^2 - 24x - 77) = 0$$

$$x = 0 \text{ and } 5x^2 - 24x - 77 = 0$$

$$5x^2 - 35x + 11x - 77 = 0$$

$$5x(x - 7) + 11(x - 7) = 0$$

$$(5x + 11)(x - 7) = 0$$

Roots of this equation are 0, 7 and $-11/5$.

II: $7y^2 + 61y + 54 = 0$

$$7y^2 + 7y + 54y + 54 = 0$$

$$7y(y + 1) + 54(y + 1) = 0$$

$$(7y + 54)(y + 1) = 0$$

Roots of this equation are $-54/7$ and -1 .

III: $2z^2 - 9z - 45 = 0$

$$2z^2 + 6z - 15z - 45 = 0$$

$$2z(z + 3) - 15(z + 3) = 0$$

$$(2z - 15)(z + 3) = 0$$

Roots of this equation are $15/2$ and -3 .

Hence the answer is option C.

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